

Optimization and Applications

MATH 145 – Applied Calculus

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Global Maximum and Global Minimum

Last lecture, we discussed how to find *local* (or *relative*) extrema.

What if we want the **global extrema** (or **absolute extrema**) of a function?

This is usually called **optimizing a function**.

Global Maximum and Global Minimum

Global Maximum

If $f(p) \geq f(x)$ for all x in the domain of f , the **global maximum** of f occurs at p .

Global Minimum

If $f(p) \leq f(x)$ for all x in the domain of f , the **global minimum** of f occurs at p .

Warning: Whether or not either of these (or both) exists depends of the *domain* of the function.

The Extreme Value Theorem

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If $f(x)$ is continuous on a closed and bounded interval, then $f(x)$ has a global maximum *and* a global minimum.

The question, of course, is where do these extrema occur?

The Extreme Value Theorem

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If $f(x, y)$ is continuous on a closed and bounded interval in the plane, then $f(x, y)$ has a global maximum *and* a global minimum.

The question, of course, is where do these extrema occur?

Theorem

If $f(x)$ is a continuous function on an interval $a \leq x \leq b$, then the global extrema occur at either:

1. The endpoints a or b .
2. The critical points of $f(x)$ inside the interval $a < x < b$.

Therefore, to find the *global extrema* of a function $f(x)$, you only need to check the endpoints of the interval and the critical points within the interval.

Example: Optimization

Example: Find the *global* maximum and minimum of the function $f(x) = x^4 - 8x^2$ over the interval $[-3, 1]$.

Example: Optimization

Example: Find the *global* maximum and minimum of the function $g(t) = \sin(t) + \cos(t)$ over the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Example: Optimization

Example: Find the *global* maximum and minimum of the function $h(x) = xe^{-x}$ over the interval $[0, 3]$.

Rule for Extrema on Open Intervals

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If $f(x)$ is continuous on an open interval and:

1. $f(x)$ has **only one** critical point,
2. $f'(x)$ changes sign at the critical point,

then that critical point is a global extrema – with type determined by the first derivative test.

Example: Optimization

Example: If possible, find the global maximum and minimum of the function $f(x) = x + \frac{1}{x}$ for $x > 0$.

Example: Constrained Optimization

Example: Find the dimensions for a closed rectangular box with a square base which has minimal surface area and a fixed volume of 8 cm^3 .

Example: Constrained Optimization

Example: A rectangle is inscribed with its base on the x -axis and its upper corners on the parabola $y = 1 - x^2$. What are the dimensions of such a rectangle with the greatest possible area?

Example: Constrained Optimization

Example: A rectangular box with a square bottom and closed top is to be made from two materials. The material for the sides costs \$1.50 per square foot and the material for the top and bottom costs \$3.00 per square foot. If you are willing to spend \$15 on the box, what is the largest volume it can contain?